

Golden Test Cases

This document provides concrete examples to configure the AI's tone, guardrails, and overall precision. It covers each knowledge category defined in the project proposal.

Example 1

- User Prompt: "I have a new customer on the phone who wants an emergency repair. They don't have a credit account with us. What are the steps to book this job?"
- Expected AI Behavior: The AI recognizes this as a "Cash Account" process and provides the exact sequence required before any tech is dispatched.
- Expected Response:
 - "🔒 **CASH ACCOUNT PROTOCOL:** Follow these steps before booking the job:
 - **Do Not Dispatch Yet:** No technician can be sent out until payment terms are secured.
 - **Create Temporary Profile:** Open BMS and generate a work order to generate the required invoicing amount.
 - **Collect Pre-Authorization:** Request a credit card deposit covering the diagnostic fee + estimated travel time.
 - **BMS Note:** Log the transaction in the Order Entry on BMS. Once approved, you may proceed to book the dispatch in FieldAware."

Example 2

- User Prompt: "Hi, Tech David started his shift at 5 AM today and has been working continuously since. It is now 4:30 PM. Can I dispatch him to a nearby job that is urgent and will take 3 hours to complete?"
- Expected AI Behavior: The AI must perform a mathematical check against the 14-hour rule and issue a strict non-compliance block.
- Expected Response:
 - "⚠️ **COMPLIANCE ALERT:** You cannot assign this job to Tech Dave today.
 - **Current Status:** He has already logged 11.5 hours (5:00 AM to 4:30 PM).
 - **Impact:** A 3-hour job will push him to 14.5 total hours, violating the strict **14-hour daily work limit**.
 - **Action Required:** Please select a different technician or schedule this job for tomorrow morning."

Example 3

- User Prompt: "We have a service call from a marina where a Cummins generator is located inside a boat. Who should I send for the job?"
- Expected AI Behavior: The AI references the static skill matrix and technician history, and suggests the correct personnel.
- Expected Response:
 - "🚢 **MARINE SKILL MATCH:** Based on our capability matrix and technician history, marine require certain attributes.
 - **Primary Suggestion:** Tech Mike (Serviced boats in marinas before and currently free).
 - **Backup Suggestion:** Tech PL (Smaller frame, but currently scheduled for a service call).
 - **Do Not Send:** Tech Ben or Tech Seb (larger-than-average frames, won't fit in the boats)."

Example 4

- User Prompt: "The client is seeking formal advice from me (service coordinator) regarding electrical work that might be more appropriate for an electrician."
- Expected AI Behavior: The AI triggers a "Do not give advice" based on department unwritten rules to prevent financial risk.
- Expected Response:
 - "🚫 **DEPARTMENT RULE:** Coordinators are **never** permitted to give formal advice to clients regarding matters that could make Cummins financially liable.
 - **Reason:** On-site variables often lead to differences in expectation and reality, which may cause financial burdens to Cummins.
 - **Correct Workflow:** Instruct the client to refer himself to an expert electrician or we send out a technician to diagnose but he will have to pay for the service."